

Baseline vs Alternative Training Recipe

Same TinyLeafGate student · same leaf / not_leaf dataset · different training process

This report compares the **baseline training recipe** (*aclis_leaf_gate_train_tflite.ipynb*, Colab run 2, 2026-07-28) with the **alternative recipe** (*aclis_leaf_gate_alt_distill_qat.ipynb*, Colab capture 2026-07-29). Architecture, input size, class set, and test split are held fixed. The only intended difference is how the student is trained and exported to INT8. That is the comparison that matters for deciding which recipe to flash onto the STM32F746.

Sources used

Baseline: Leaf Gate Model/Baseline Leaf Gate Model/Colab Training Run History/TRAINING_RESULTS.md, results.pdf, run-2 notebook.
Alt: Leaf Gate Model/Alt Leaf Gate Model/ALT_TRAINING_REPORT.md, alt_training_record.pdf, results.pdf. Side-by-side INT8 error counts (including baseline-under-stress) come from alt report §3.2, which evaluated both TFLite artifacts on the same clean and camera-stress images.

1. What stayed the same

Both recipes deploy the same MCU-sized depthwise-separable CNN (TinyLeafGate: stem + 4 DW-separable blocks + GAP + 1×1 classifier, 28,854 parameters). Both take 96×96 RGB, both decide leaf vs not_leaf before the PlantVillage disease CNN, and both export through the same Keras twin / TinyEngine INT8 path. Flash cost is essentially identical (50.9 KB vs 51.1 KB). Isolating the student means any gain or loss is attributable to the recipe, not a bigger net.

Student	TinyLeafGate, 28,854 params	Same topology
Task / cascade	leaf / not_leaf → disease CNN	Same
Input	96×96 RGB, full-frame Resize(96)	Same
Dataset	Kaggle leaf / not_leaf, n_test = 3,673	Same split
Train / val / test	17,131 / 3,670 / 3,673	Same counts
Export path	Keras twin → full INT8 TFLite	Same TinyEngine path
INT8 size	50.9 KB	51.1 KB (+0.2 KB)
Disease + gate flash	773.9 / 1024 KB (FITS)	774.1 / 1024 KB (FITS)

Table 1. Architecture, data, and MCU budget are shared.

2. What the recipes change

Baseline is a two-phase Adam schedule with geometric and colour jitter, then post-training quantisation. Alt adds a MobileNetV3-Small teacher, knowledge distillation, MixUp, EMA, camera-like degradations (blur, JPEG, harsher lighting), a single warmup–cosine AdamW schedule, and a short INT8-noise QAT stage before the same PTQ export.

Teacher / KD	None — hard labels only	MobileNetV3-Small + KL ($\alpha=0.7$, T=4)
Student epochs	5 warm-up + 25 fine-tune	40 KD (+ 5 QAT proxy)
Optimiser	Adam, lr 1e-3 then 1e-4	AdamW warmup–cosine, peak 1e-3
Regularisation	Class weights + sampler	+ MixUp $\alpha=0.2$, label smooth 0.05, EMA 0.999
Train augs	Crop / flip / rotate / jitter / erase	+ Gaussian blur, JPEG q=35–90, stronger light
Quantisation	PTQ only	INT8-noise QAT (5 ep) then PTQ

Eval protocol	Clean test	Clean + fixed camera-stress + vs baseline TFLite
Notebook target	≥ 92% (PASS)	≥ 92% (PASS)

Table 2. Alt invests complexity at train time only. The deployed graph stays TinyLeafGate.

3. Overall accuracy

On the clean Kaggle test set both models are already near ceiling and essentially tied. Baseline INT8 is 99.13%; alt INT8 is 99.07% (−0.05 pts). The recipe is judged on the **camera-stress** split — the same 3,673 images with a fixed blur + underexposure + JPEG q=55 pipeline, meant to stand in for OV2640 / Arducam frames. There baseline drops 1.50 points to 97.63%; alt drops only 0.21 points to 98.86%. That +1.23-point stress gap is the headline result.

Baseline clean INT8 99.13% n = 3,673 · PTQ	Alt clean INT8 99.07% −0.05 pts vs baseline	Baseline stress INT8 97.63% drop 1.50 pts	Alt stress INT8 98.86% +1.23 pts vs baseline
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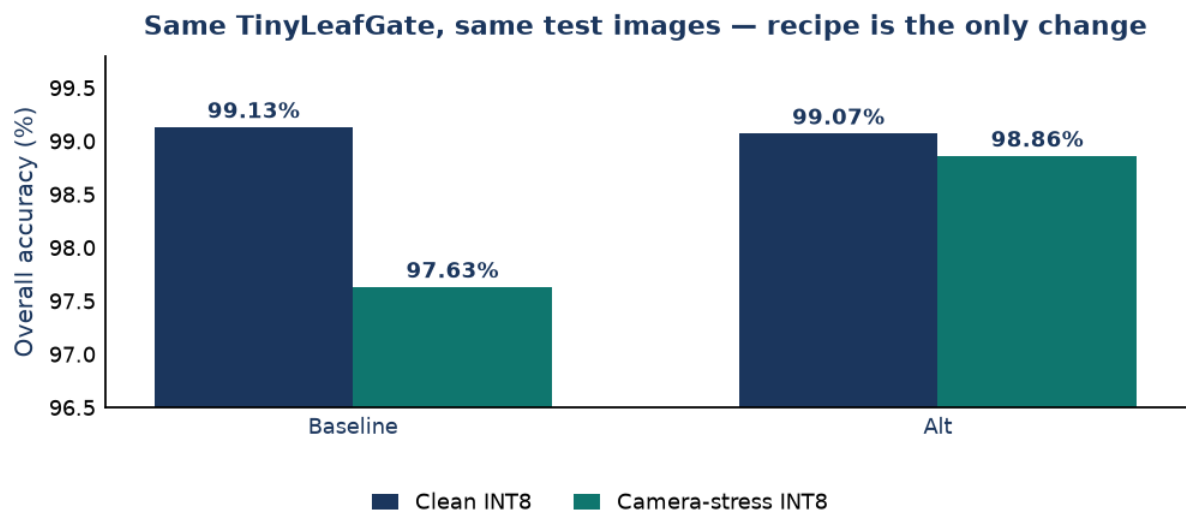


Figure 1. Clean scores are saturated. The recipes separate under camera-like degradation.

Clean INT8 overall	99.13%	99.07%	Tied on Kaggle; both ■ 92% target
Clean PyTorch FP32	99.10%	~99.1–99.5%	Quantisation is not the difference
INT8 – FP32	0.03 pts	~0.03 pts	Both pass the 1% export gate
Best validation	99.13% (ckpt)	99.10% (EMA)	Student selection metric
Camera-stress INT8	97.63%	98.86%	Alt preferred for field proxy
Clean → stress drop	1.50 pts	0.21 pts	Main evidence the augs closed the gap

Table 3. Overall numbers from each folder's INT8 test. Stress uses identical degraded images.

4. Per-class accuracy and cascade errors

Per-class accuracy is recall. FN_leaf (true leaf predicted not_leaf) skips the disease CNN. FP_leaf (true not_leaf predicted leaf) wastes an ~800 ms disease-CNN call on junk. On clean Kaggle data alt is a hair weaker on leaves (15 vs 11 false rejects) and a hair stronger on non-leaves (19 vs 21 false accepts). Under camera stress the ranking reverses: alt keeps more

real leaves on the disease path and cuts junk accepts from 70 to 32.

leaf · clean	99.43% (1,932/1,943)	99.23% (1,928/1,943)	4 more FN	Missed leaf never reaches disease CNN
not_leaf · clean	98.79% (1,709/1,730)	98.90% (1,711/1,730)	2 fewer FP	Clutter sent into disease CNN
Overall · clean	99.13%	99.07%	−0.05 pts	All 3,673 test images
leaf · stress	99.13% (1,926/1,943)	99.49% (1,933/1,943)	7 fewer FN	Alt keeps more leaves under blur/JPEG
not_leaf · stress	95.95% (1,660/1,730)	98.15% (1,698/1,730)	38 fewer FP	Largest field win: fewer junk disease calls
Overall · stress	97.63%	98.86%	+1.23 pts	Decisive split for OV2640 / outdoor use

Table 4. Recall and error counts. Positive Δ on stress means alt is better for the board camera.

5. Confusion matrices (clean INT8)

Rows are true labels; columns are predictions. Both models make only a handful of leaf $\leftrightarrow$ not_leaf swaps on clean Kaggle data (32 mistakes for baseline, 34 for alt). Those extra two alt errors are the 0.05-point clean regression — not a class collapse.

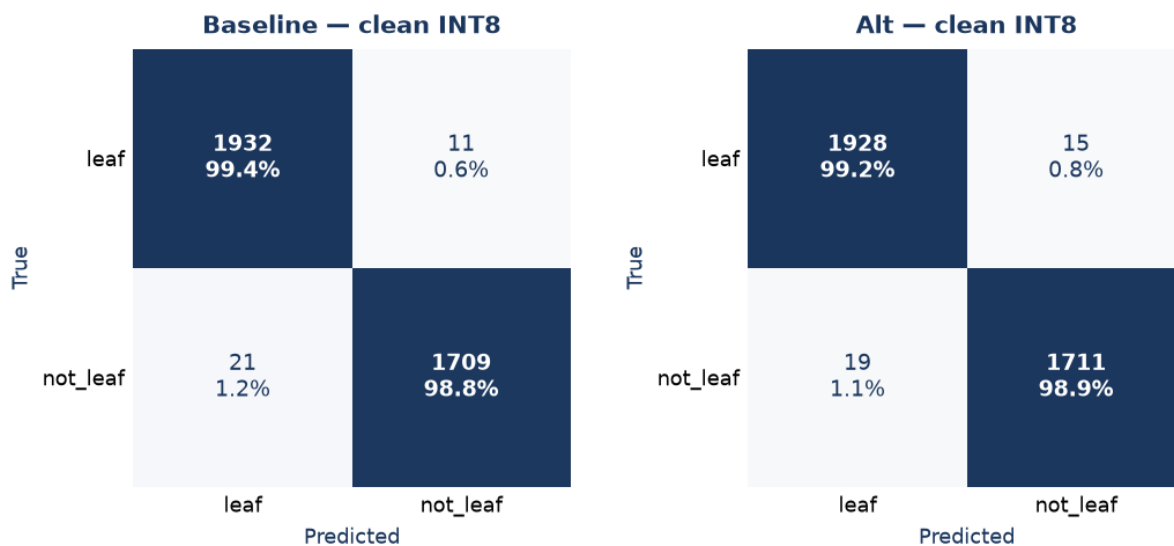


Figure 2. Clean INT8 matrices reconstructed from recorded FN_leaf / FP_leaf and test-set sizes.

leaf → leaf	1,932	1,928	Disease CNN runs (desired for a real leaf)
leaf → not_leaf (FN)	11	15	Frame rejected; disease CNN skipped
not_leaf → leaf (FP)	21	19	Junk frame enters disease CNN
not_leaf → not_leaf	1,709	1,711	Reject clutter (desired)

Table 5. Shared error types on the clean test set.

6. Why alt should beat baseline in the field

Baseline already saturates clean Kaggle test. Extra points there are mostly noise. The ACLIS failure mode that matters is the board camera: soft focus, JPEG from the Arducam path, uneven outdoor light, and leaves that do not fill the frame. Alt trains on those degradations and then measures both TFLite artifacts on a fixed stress set.

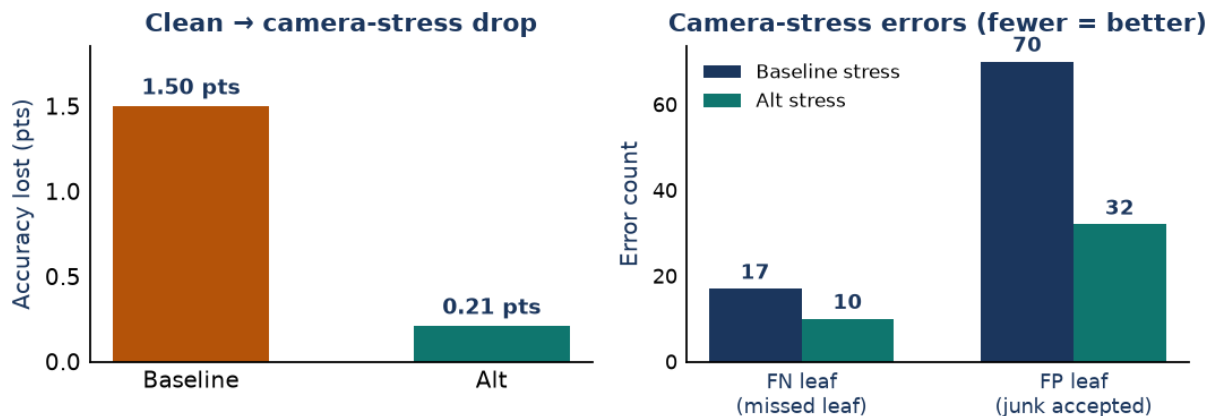


Figure 3. Alt barely moves under stress. Baseline’s extra errors are mostly false leaves (FP 21 → 70).

Field-matched augmentation is the main lever: blur ($p=0.35$), JPEG ($p=0.4$), and harsher colour/lighting during training. **Knowledge distillation** gives the tiny student soft targets from a ~1.52 M teacher that is never deployed. **MixUp + EMA** soften overconfident boundaries on ambiguous frames. **QAT** keeps INT8 within ~0.03 pts of FP32, same as baseline’s PTQ gap — so the stress win is not a quantisation story. What alt does *not* claim: a clean-Kaggle win. The -0.05 pt clean dip and +4 FN_leaf are the expected trade for out-of-distribution robustness.

7. Conclusion

Use the **alternative recipe** when the gate will see STM32 + OV2640 / outdoor frames. It keeps the same 51 KB TinyLeafGate, stays within 0.05 points of baseline on clean Kaggle data, and is clearly better on the camera-stress proxy (+1.23 pts overall, 7 fewer missed leaves, 38 fewer junk disease-CNN calls). Flash `aclis_leaf_gate_96x_alt_full_int8.tflite` via `codegen_leaf_gate_alt_c.py`.

Keep the **baseline recipe** as the fallback if on-device clean behaviour ever regresses. Its 5+25 Adam schedule is simpler, it is 0.05 points ahead on in-distribution Kaggle test, and `codegen_leaf_gate_c.py` can restore those weights. A field A/B should score two things on real captures: (1) true leaves still entering the disease CNN, (2) clutter not wasting a disease-CNN call.

Highest clean Kaggle INT8	Baseline (99.13% vs 99.07%)
Robustness to blur / JPEG / lighting	Alt (98.86% vs 97.63% stress)
Fewer missed leaves under camera stress	Alt (FN 10 vs 17)
Fewer junk frames into the disease CNN under stress	Alt (FP 32 vs 70)
MCU flash / SRAM	Either (~51 KB; both fit with the 723 KB disease model)

Table 6. Decision guide. The notebooks’ own recommendation from this table is to prefer alt for field use.